

EEET306

EE ELECTRONICS

Module 1

are the performance of various power semiconductor devices



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Module 1

Electronics-Scope and applications-power electronics vs signal

of operation of power devices- Power diode, Power MOSFET & characteristics - comparison. Basic principles of wideband gap devices- characteristics & Switching (turn-on & turn-off) characteristics - turn-on methods of SCR - Two transistor analogy (5 hr)

– Requirements of isolation and synchronization in gate drive transformer based isolation (1hr)

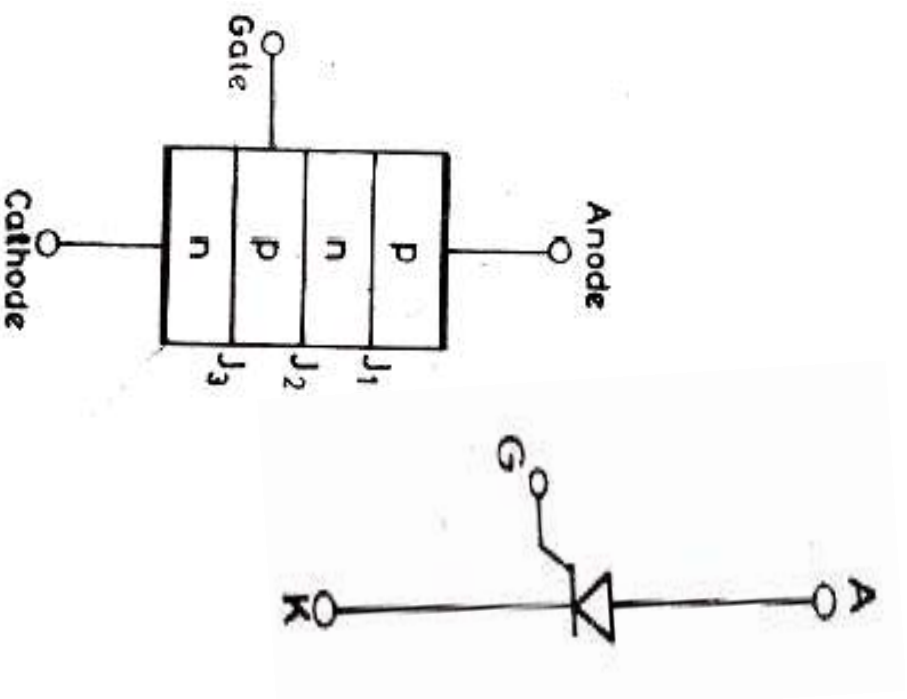
Controlled Rectifier (SCR)

action, p-n-p-n
or switching device.

Anode, Cathode,

l device

ed ON by means of
, but can't be



SCR Characteristics

Characteristics: Relationship between Anode

Cathode Voltage (V_{ak}) and Anode Current (I_a)

Switching/Transient Characteristics: Time

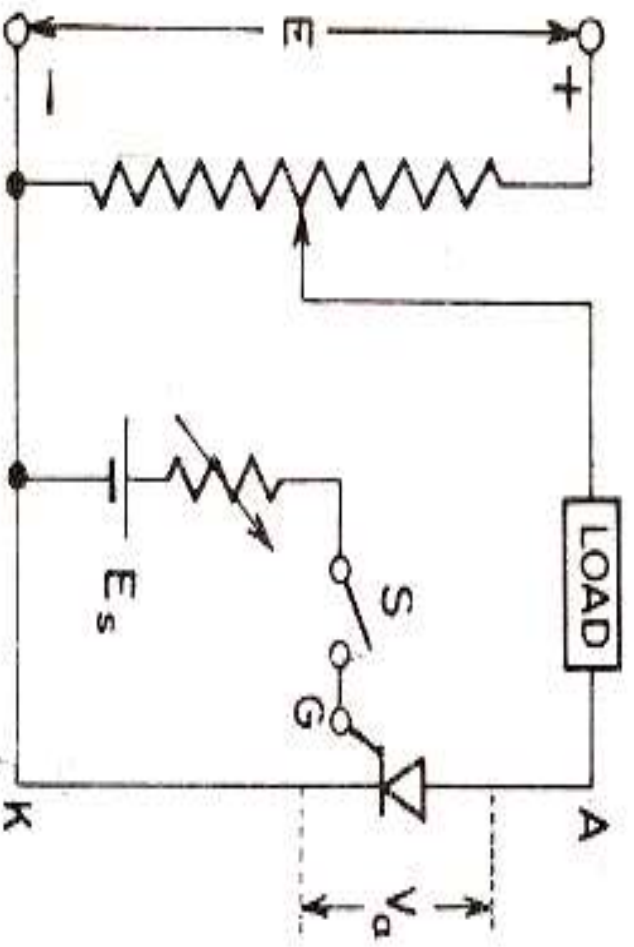
of the voltage across thyristor and
through it during turn on and turn off

Characteristics of SCR

g Mode

ng Mode

ction Mode



Characteristics of SCR (cont..)

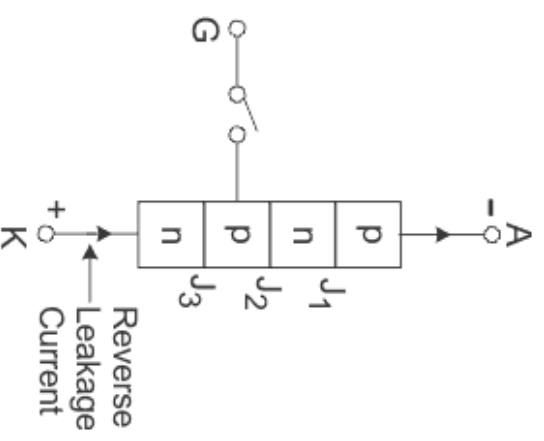
3 Mode

positive with respect to anode by

the supply voltage E_s is detached
with S open.

are reverse biased whereas the
J is biased.

all leakage current of the order of a
this current is called reverse leakage



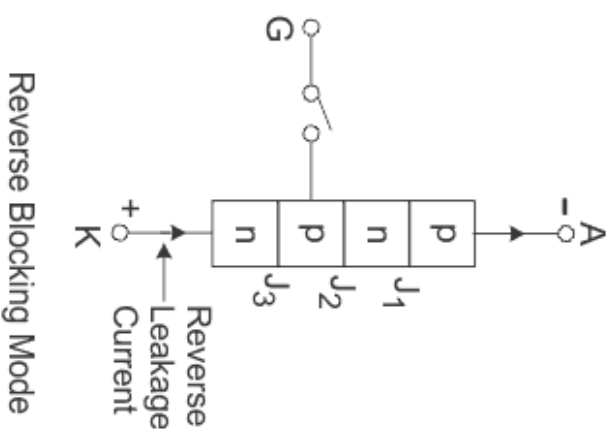
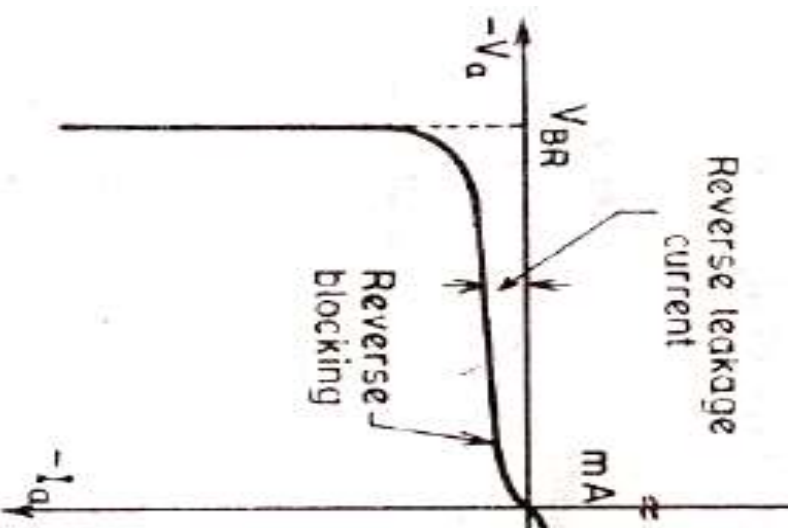
Reverse Blocking Mode

Characteristics of SCR (cont..)

3 Mode

blocking
e, of the

e is now
at a
nown as
voltage
ars at the
ctions J_1



Characteristics of SCR (cont..)

Forward Mode

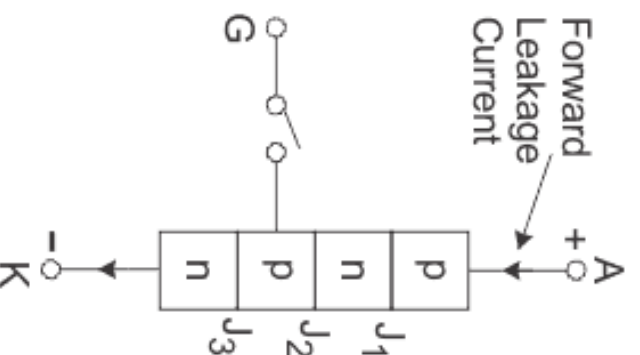
Forward Mode: When the SCR is forward biased (Anode positive with respect to the Cathode) and a voltage E is applied across it, it is in the forward blocking condition.

In this condition, the SCR is forward biased but it does not conduct. It is in the forward blocking condition.

When the forward bias is increased, a small current, called forward leakage current, is allowed to flow.

This current is called *forward leakage current*.

The SCR remains in the forward blocking mode and does not switch to the ON state (open switch).

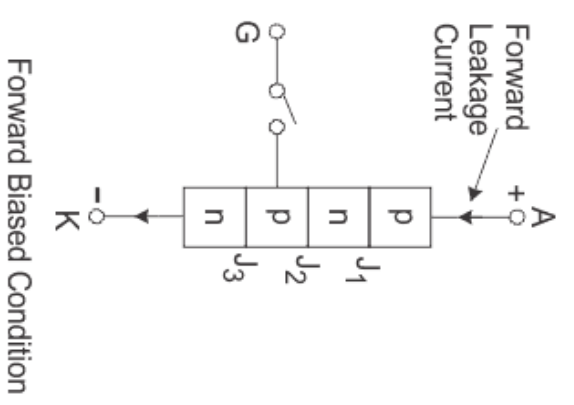
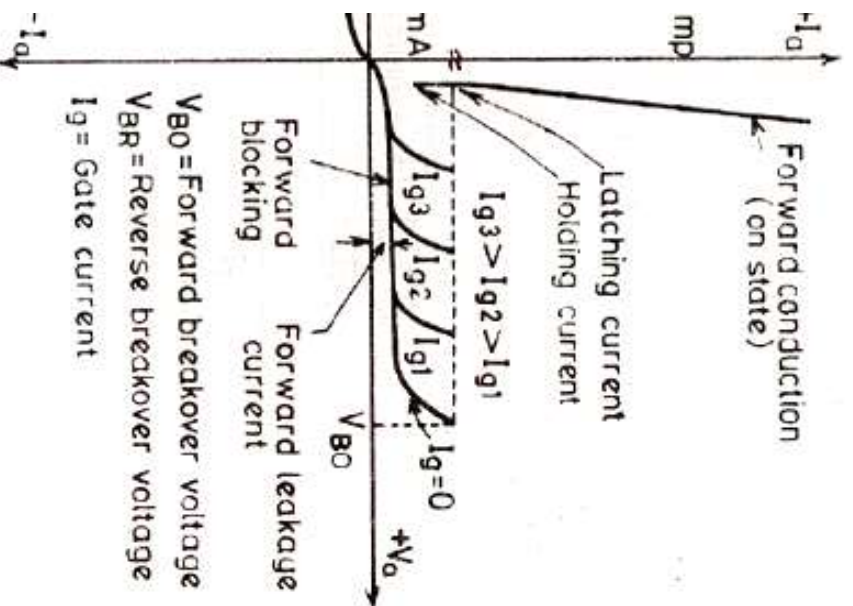


Forward Biased Condition

Characteristics of SCR (cont..)

g Mode

ge is now
at a
known as
voltage
occurs at
junctions



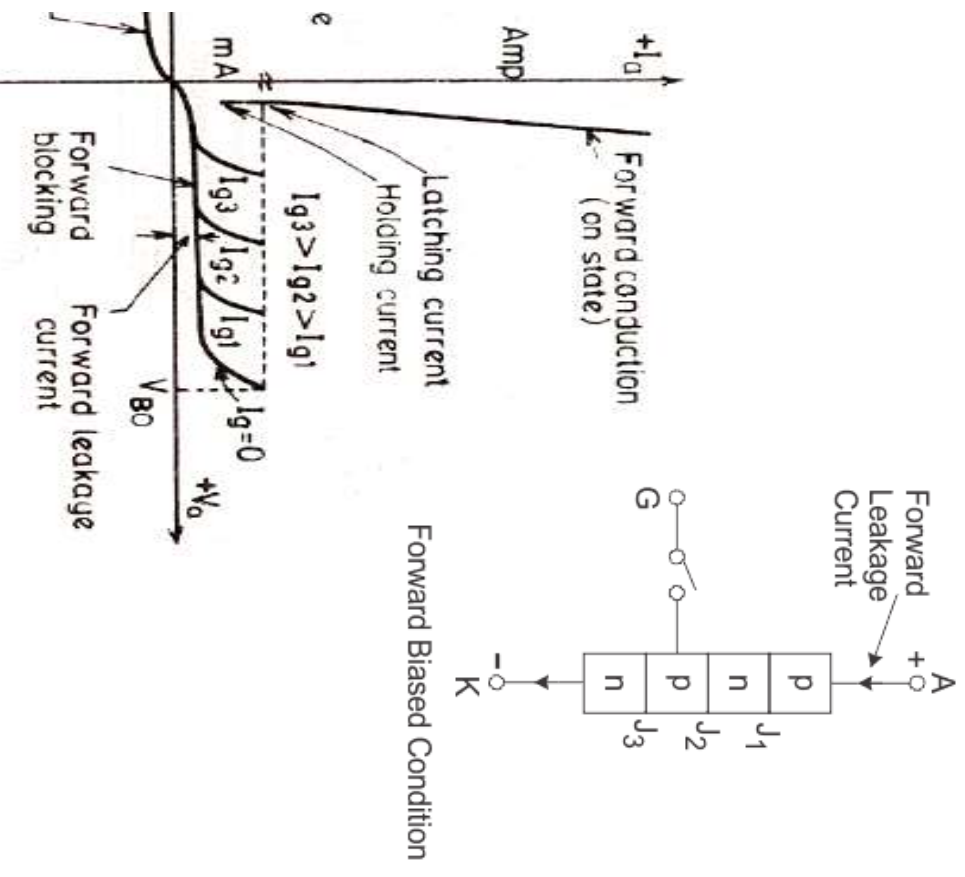
Characteristics of SCR (cont..)

Forward Mode

With cathode forward biased, with gate circuit action J_2 will have breakdown at forward voltage V_{BO} leading to

conducts maximum current voltage drop. controlled by the

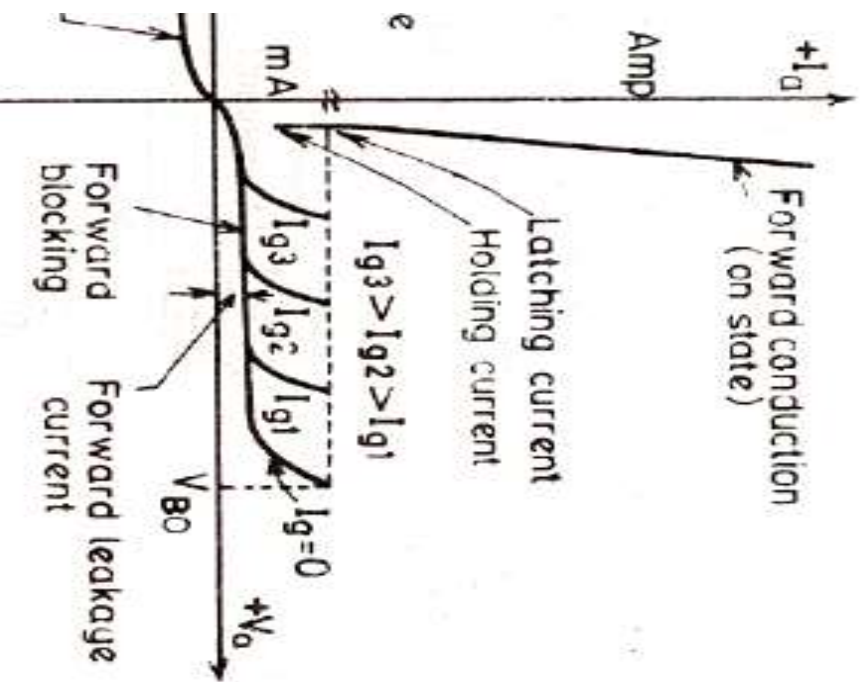
forward conduction on mode of the



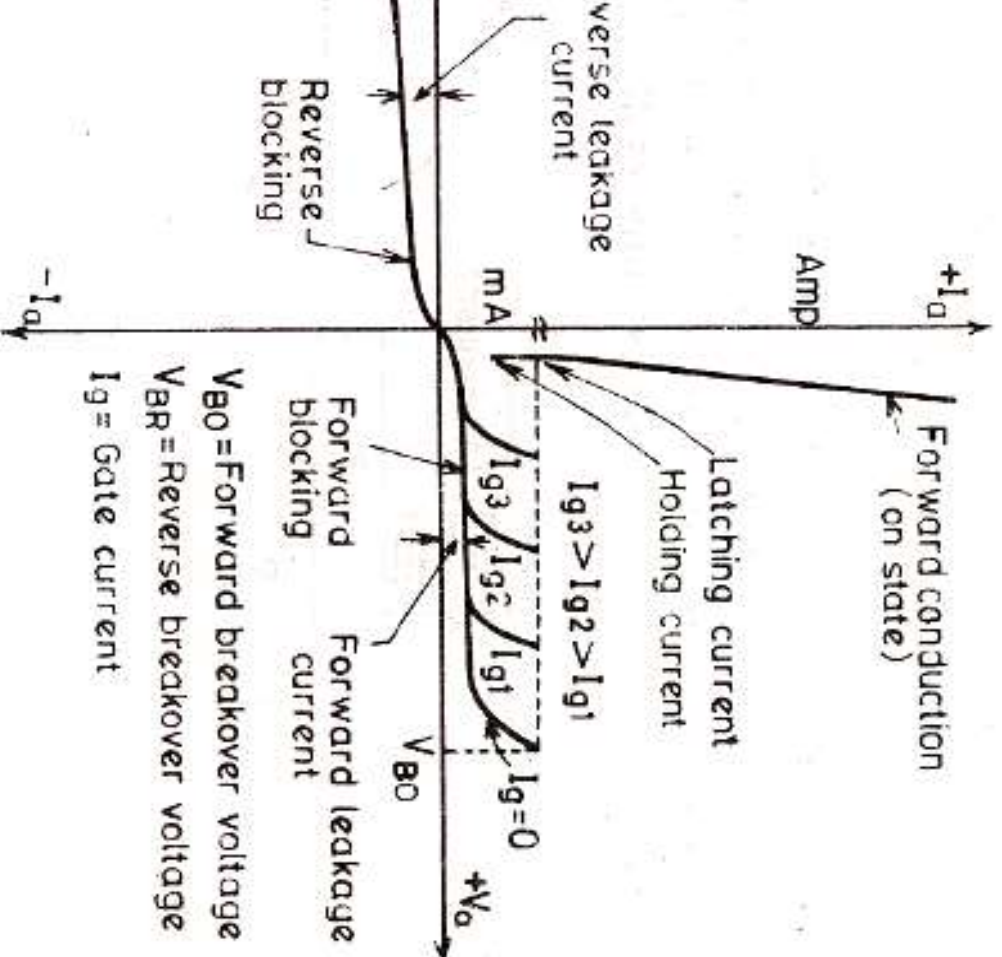
Characteristics of SCR (cont..)

Conduction Mode

When a positive voltage is applied across gate and anode, the switch S, the forward bias across the junction J_2 will happen at V_{BO} . As the gate current, I_g , is applied, the conduction will happen at a lower voltage V_{BO} .



Characteristics of SCR (cont..)



g Current & Holding Current

es from forward-blocking to forward conduction, the
low to flow a sufficient anode current to keep the device

**ode current to maintain the SCR in ON state even though
ved.**

eady in forward conduction and the anode current is
e can move its operating mode from forward-
forward-blocking.

**ode current necessary to keep the device in forward
ch SCR falls to OFF state.**

g Current & Holding Current

urrent value is lower than the latching

minimum current required to start conduction.

minimum current required to maintain

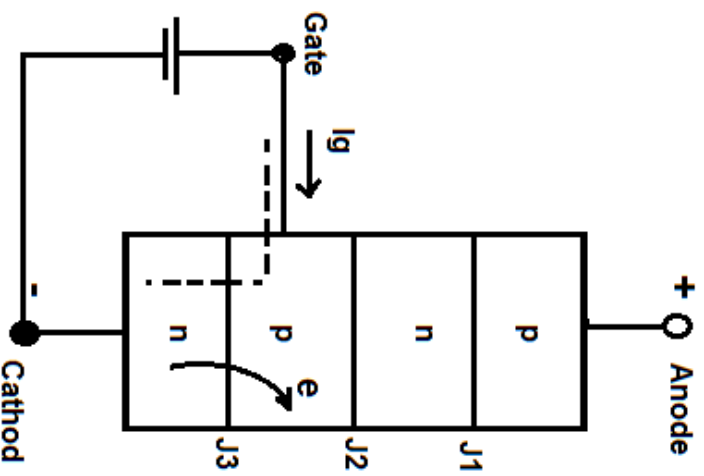
– ON Methods of SCR

Triggering: A forward voltage is applied across the terminals with gate terminal open. As forward bias is applied, the junction J1 and J3 are forward biased above forward break over voltage, avalanche occurs at junction J2. As other junctions J1 and J3 are forward biased, breakdown of J2 allows free movement of electrons from the junction and as a result, large forward anode current flows.

Discuss the concept of various power semiconductor devices

– ON Methods of SCR

Turning on of SCR by gate
Simple, reliable and efficient,
the most usual method of
turning on the SCR
is applied across gate and
cathode. Thus SCR can be turned
on at a voltage less than the
breakover voltage.



– ON Methods of SCR

dv/dt Triggering is the technique in which SCR is turned ON forward bias voltage with respect to time. dv/dt itself means voltage w.r.t time.

Reversed biased in a forward blocking mode of SCR. A reversed biased junction may be treated as a capacitor due to presence of space charge in the vicinity of reversed biased junction.

$$i = C \frac{dv}{dt}$$

When the SCR is forward biased, the rate of rise of forward voltage dv/dt is directly proportional to the rate of rise of forward current di/dt . Therefore if the rate of rise of forward voltage i.e. dv/dt is high, the rate of rise of forward current di/dt will also be high. This charging current will turn the SCR ON even though the gate current is not present.

– ON Methods of SCR

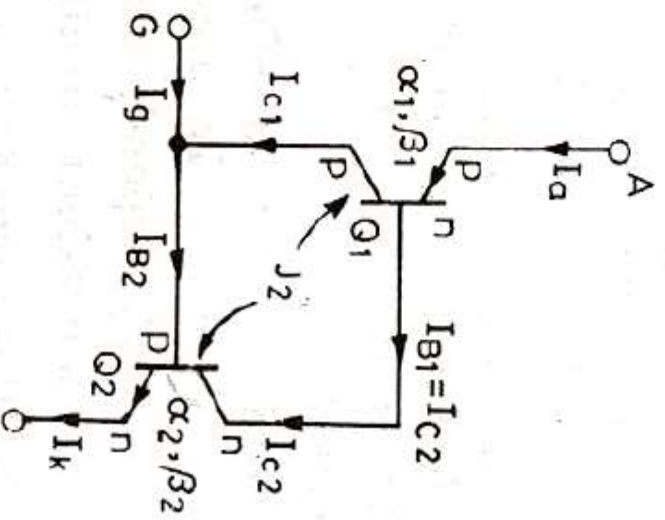
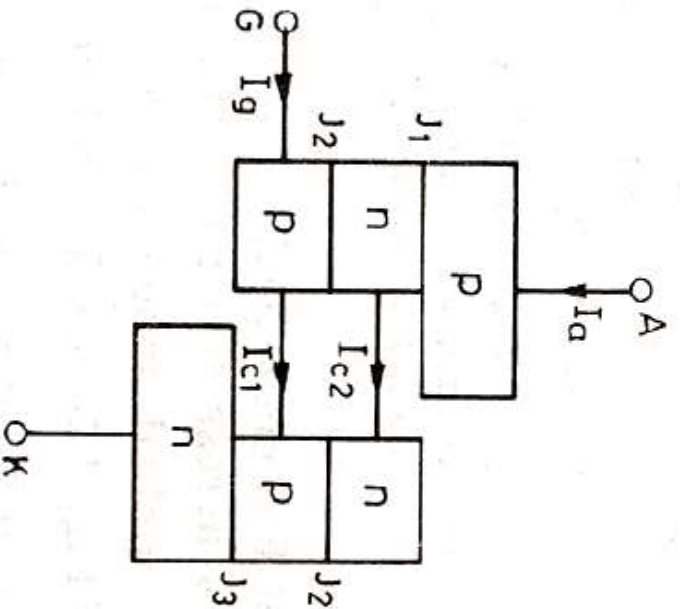
Triggering: During forward blocking, most of the appears across the reverse biased junction J2. This junction associated with leakage current may raise of this junction. With increase in temperature, further increases. This cumulative process may turn the high temperature.

This triggering is only applicable for light activated device, a pulse of light of appropriate wavelength is used to turn on the SCR. When the intensity of this light exceeds a certain level, the SCR is turned on.

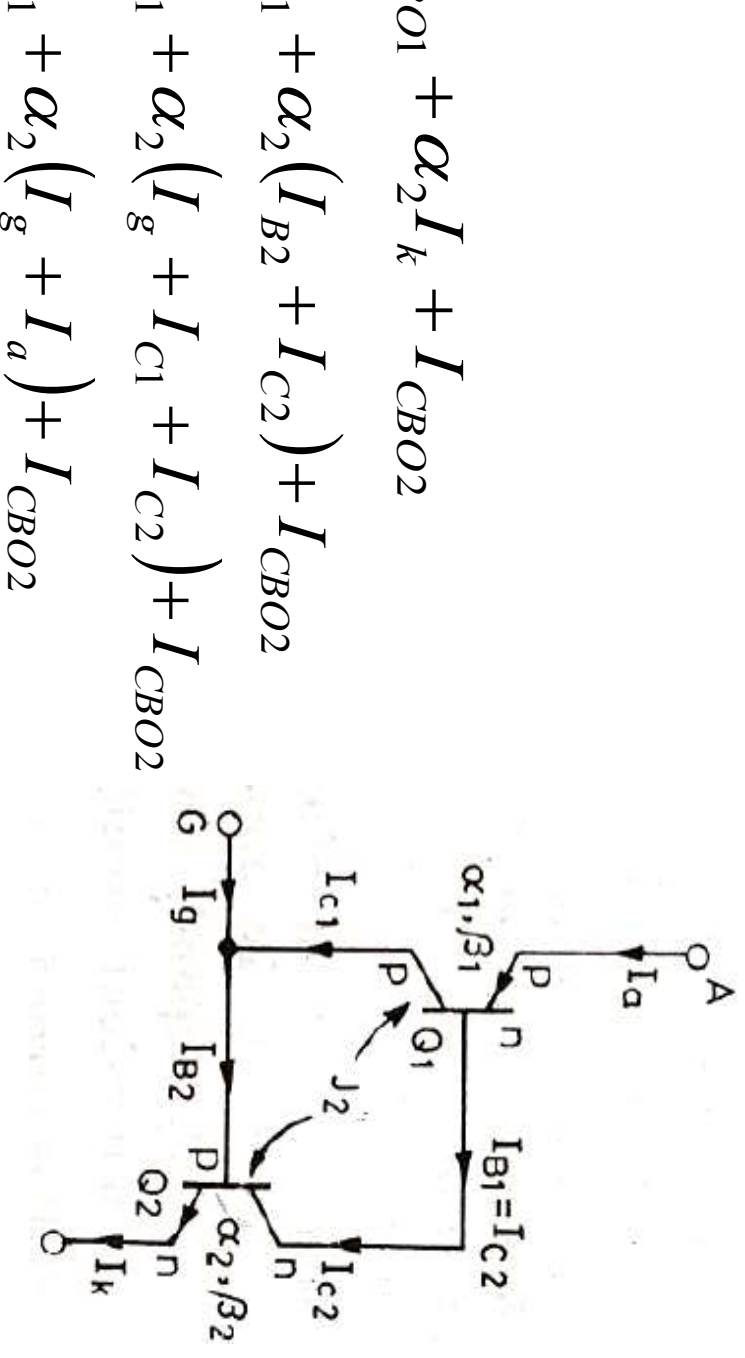
Transistor Model of SCR

principle of SCR, can be easily understood by the two or analogy) of SCR.

ough the dotted line, we will get two transistors



Resistor Model of SCR (cont..)



$$I_a = \frac{\alpha_2 I_g + I_{CB01} + I_{CB02}}{1 - (\alpha_1 + \alpha_2)}$$

Resistor Model of SCR (cont..)

$$\frac{+I_{CBO2}}{\alpha_2})$$

- During forward blocking mode, only leakage current will flow through SCR and $I_g = 0$.
- Since anode current is small, α_1 and α_2 will be small
- Thus SCR remains in OFF state.
- When I_g is applied, I_a increases and thus α_1 and α_2 will increase.
- Again I_a increases.
- This cumulative process will turn on the SCR.

I_E

Dynamic Characteristics of SCR

Dynamic Characteristics

Dynamic characteristics of the voltage across thyristor and current through it during turn on and turn off process

Turn On

Characteristics: Time variations of voltage across and current through it during turn ON process.

Characteristics: Time variations of voltage across and current through it during turn OFF process.

ON Characteristics of SCR

thyristor is turned on by applying a positive gate voltage and cathode.

on time from forward off state to forward on state.

Turn-ON time of thyristor.

ne is defined as the time during which SCR changes
cking state to final ON state.

e is divided into 3 intervals.

)

$$T_{ON} = t_d + t_r + t_p$$

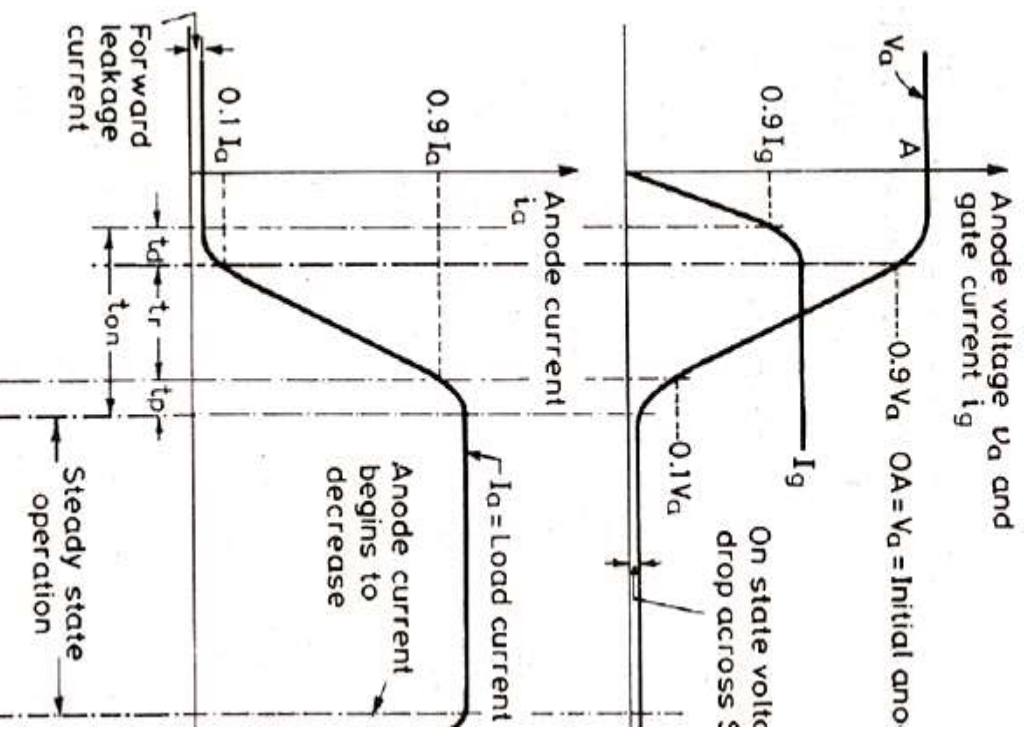
t_p)

ON Characteristics of SCR

is measured from the gate current reaches a constant at which anode current falls from V_a to $0.9V_a$. (I_g and I_a are initial values of gate and

ned as the time during
ge falls from V_a to $0.9V_a$
anode voltage)

ded to rise the anode
and leakage current to

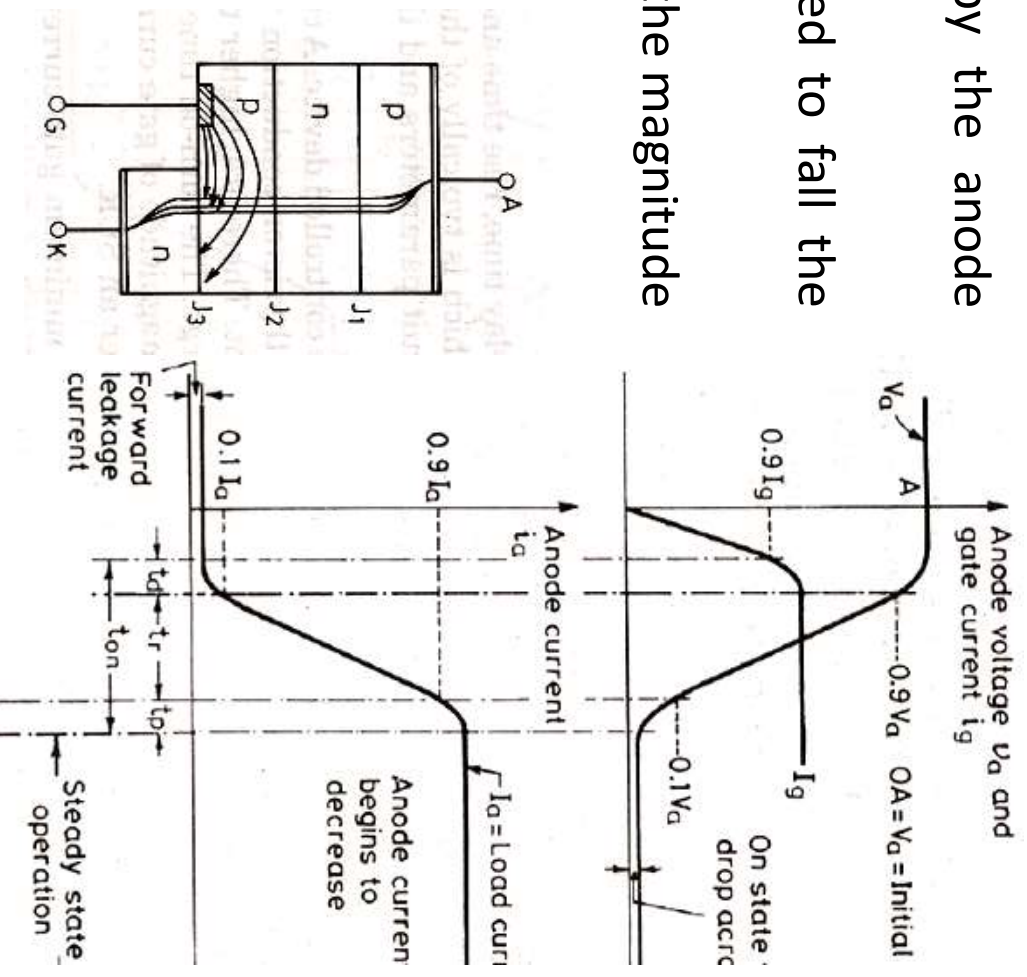


ON Characteristics of SCR

the time taken by the anode current to fall from $0.9I_a$ to $0.1I_a$ is the time needed to fall the anode current from $0.9V_a$ to $0.1V_a$ proportional to the magnitude

anode current from narrow

highest during the high anode current.



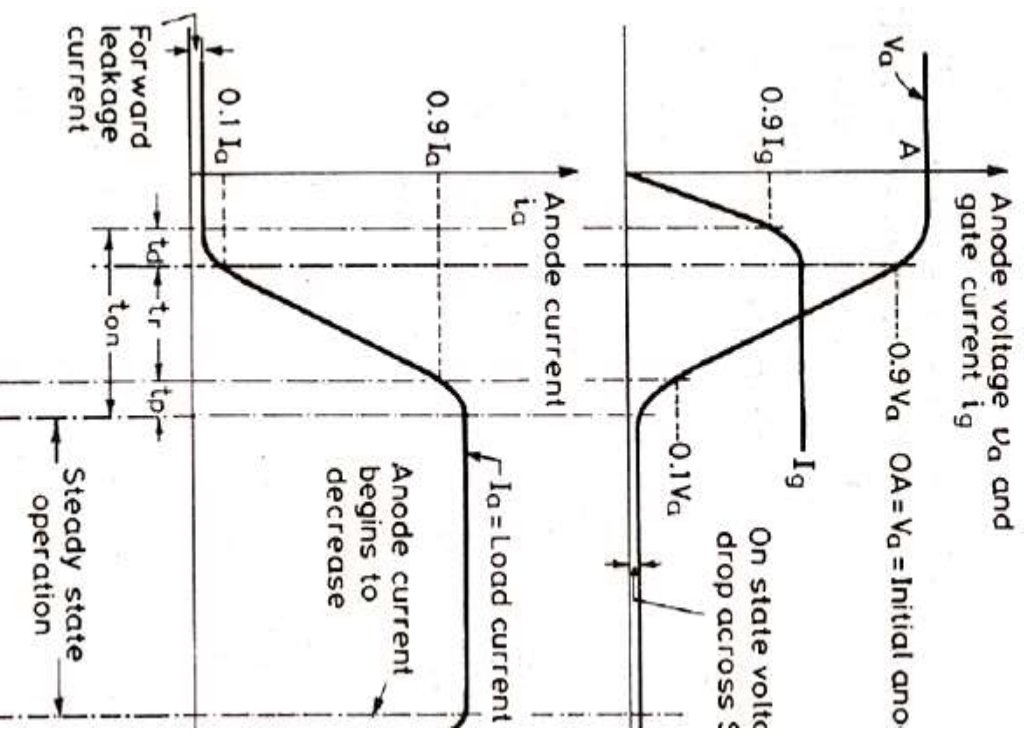
ON Characteristics of SCR

is the time taken by
to rise from $0.9I_a$ to

as the time for the
fall from $0.1V_a$ to the
op (1 to 1.5V).

conduction spreads
cross-section of the

$$+ t_r + t_p$$



FF Characteristics of SCR

cess of an SCR means the transition from the
e to the forward blocking state.

or is turned ON, gate losses the control.

rned off by reducing the anode current below

ltage is applied to the SCR at the moments its
alls to zero, the device will not be able to block this
due to the presence of charge carriers.

e will go into conduction immediately the presence
ng, even though gate signal is not applied.

FF Characteristics of SCR

ne of thyristor is defined as the time between the
rent becomes zero and the instant SCR regains forward
Y.

all the excess charge carriers from the four layers of the
ved.

define as the time required for the SCR to change from
to forward blocking state.

e is divided into 2 intervals:

very Time (t_{rr})

y Time (t_{gr})

$$T_{OFF} = t_{rr} + t_{gr}$$

FF Characteristics of SCR

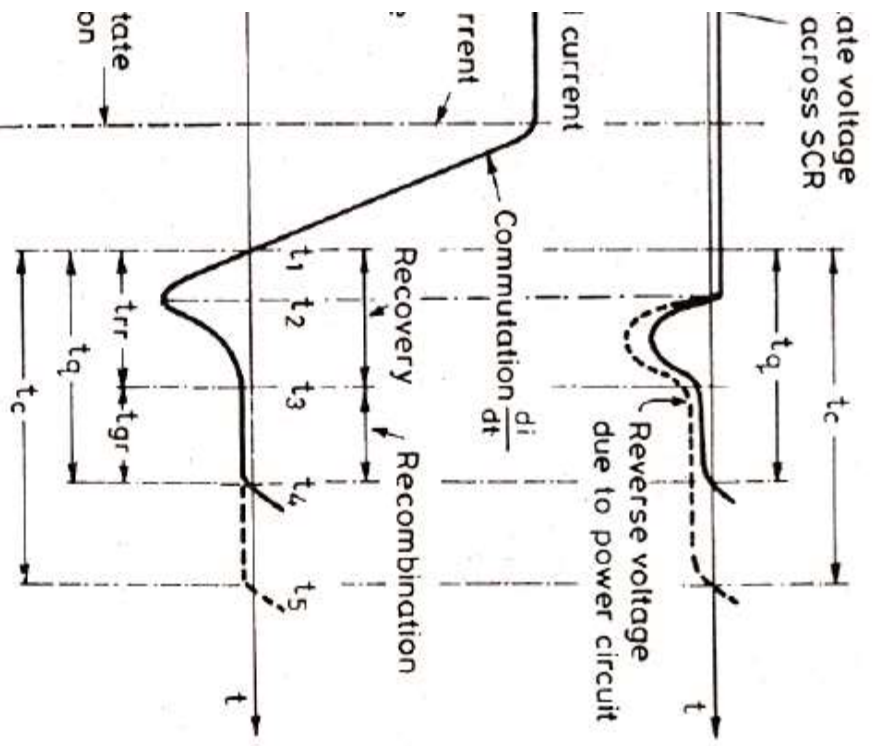
time (t_{rr}) is defined as to remove the excess junctions.

anode current reaches

rent builds up in the
th the same slope as

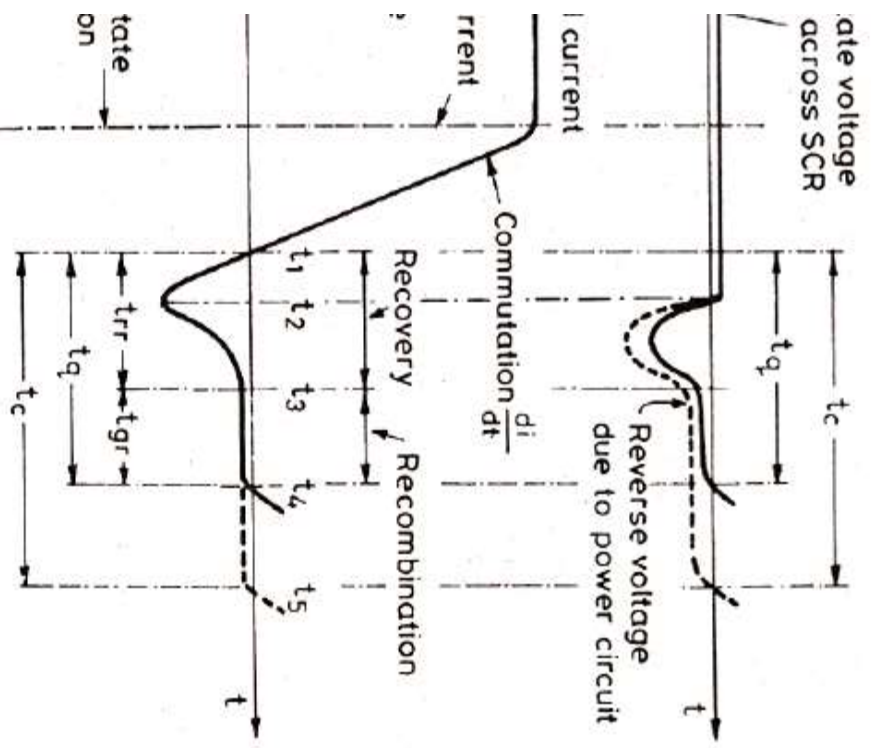
reverse current is the
in the four layers.

can remove the carriers



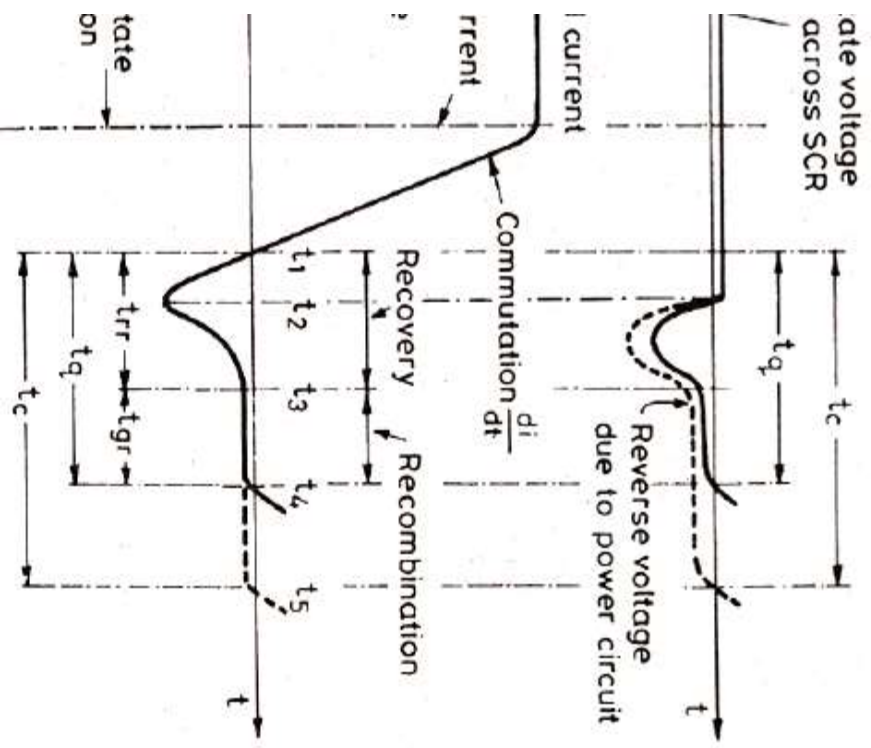
FF Characteristics of SCR

about 60% of the
e removed from the
urrent starts decaying.
urrent will cause a
oss the device due to
all the carriers are
outer layers and thus
urrent falls to zero.



FF Characteristics of SCR

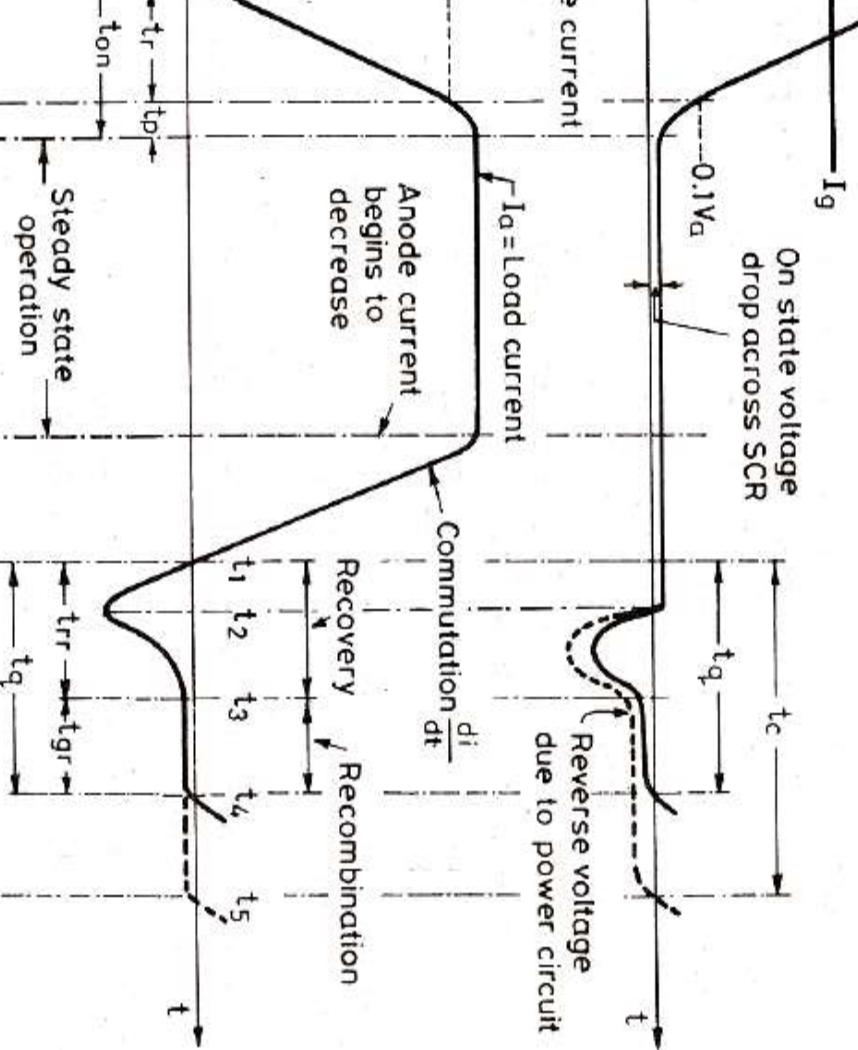
t_{gr} is defined as the time to remove the excess carriers by recombination. There will be trapped charges only by recombination. It can be possible by applying a reverse voltage across the device. When a reverse voltage is applied, the trapped charges are removed from the inner layers.



ng Characteristics of SCR

age voltage V_a and
current I_g

----- $0.9V_a$ $0A = V_a = \text{Initial anode voltage}$



THYRISTOR PROTECTION

Parameters of thyristor should not be exceeded for its safe operation.

Limitations of large di/dt and dv/dt during turning ON and OFF processes respectively.

Thyristor must be protected from all such abnormal conditions to ensure satisfactory and reliable operation.

di/dt PROTECTION

thyristor is forward biased and is turned on by a gate current, the rate of rise of anode current begins in the immediate vicinity of the gate-cathode junction.

As the current spreads across the whole area,

the rate of rise of anode current is higher than the spread rate in the immediate vicinity of the gate-cathode junction. Hot spots will be formed.

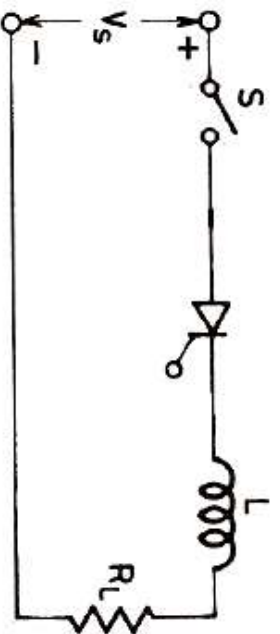
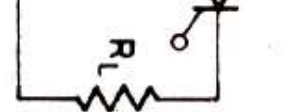
Excessive heating may destroy the thyristor.

The rate of rise of anode current at the time of turn-ON must be below the specified limit.

This can be achieved by connecting a small inductor in series with

di/dt PROTECTION

current cannot be changed instantaneously.



$\frac{dQ}{dt}$ PROTECTION

voltage across the anode and cathode of a two outer junctions are forward biased but the inner junction is reverse biased.

The two outer junctions act as a capacitor, with the entire depletion voltage across it.

A rapid change of voltage, will lead to large charging

$$i = \frac{dQ}{dt} = \frac{d}{dt} CV$$

$$= C \frac{dV}{dt}$$

$\frac{dv}{dt}$ PROTECTION

ge the thyristor.

nt plays the role of gate current and turn ON
gh there is no gate signal (false triggering).

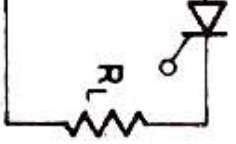
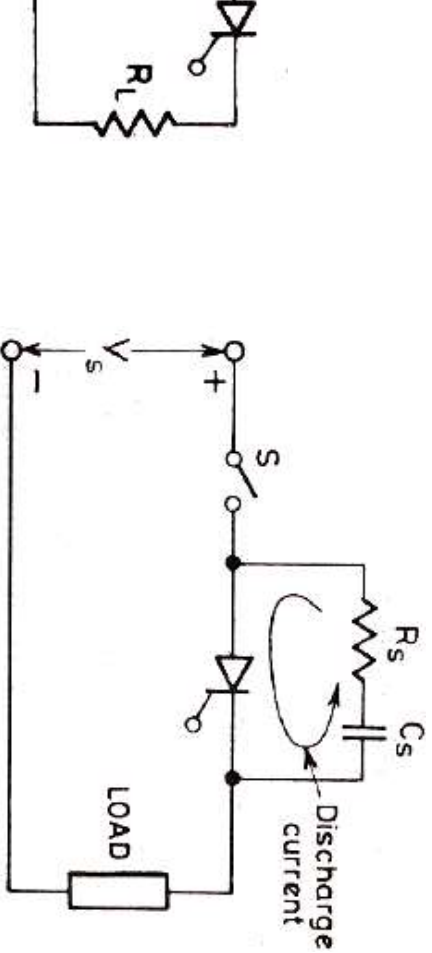
se of forward voltage should be kept in specified

ieved by connecting a capacitor in parallel with

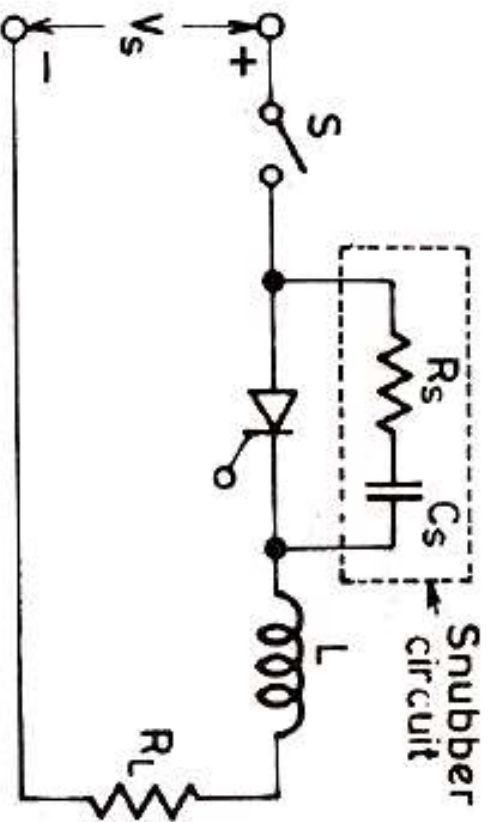
voltage cannot be changed instantaneously.

urrent during the discharging of the capacitor, a
connected in series.

$\frac{di}{dt}$ PROTECTION



di/dt & dv/dt Protection



Gate Drive Circuit

circuit is an integral part of power electronics systems.

It forms an important interface between the high-power semiconductor devices and the control circuit and are used to drive power devices.

The gate drive circuit is a power amplifier that accepts a low power input from a control circuit and produces the appropriate high current gate drive for a power semiconductor device to turn ON.

It provides isolation techniques in gate drive circuits.



on in Gate Drive Circuit

al separation between various functional circuits in a there is no direct conduction path available between

idual circuits to possess different ground potentials.

n gate drivers, isolation may be necessary for functional might also be a safety requirement.

ction against any fault on the high voltage side as the blocks electrical power from reaching the user in spite of ge or failure.

w voltage electronic equipments from any damage due gh power side.

on in Gate Drive Circuit

are two popular techniques available to implement
vers:

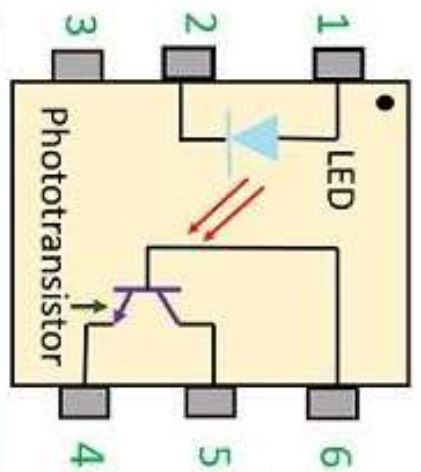
Method (using an opto-coupler – Low Frequency).

Method (using pulse transformer – High Frequency)

Opto-couplers

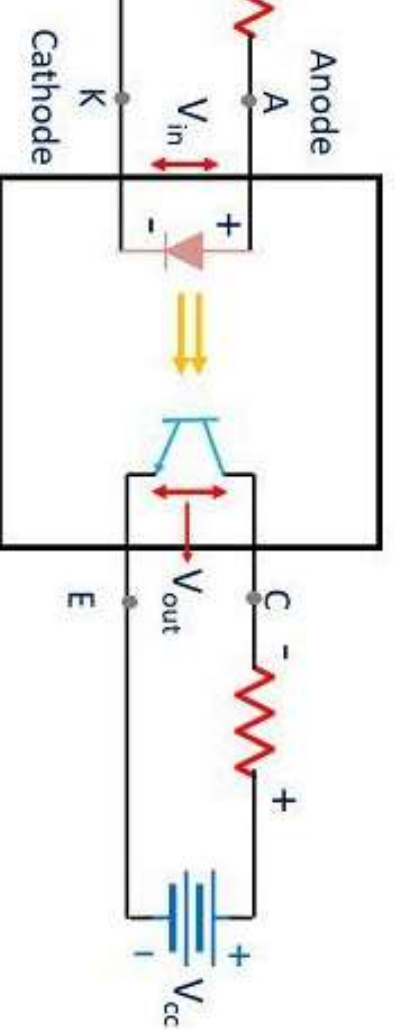
is a semiconductor device that allows an electrical signal to be transmitted between two isolated circuits.

It mainly consists of an **infrared LED** and a **phototransistor** that detects the emitted infrared beam. The output device can be a photodiode, a relay, etc.



Opto-couplers

Applied to the optocoupler, making the LED emit an amount proportional to the current going through the device. When the photosensor, it is switched on and a current starts flowing through the LED is interrupted, the IR beam causing the photosensor to stop conducting.



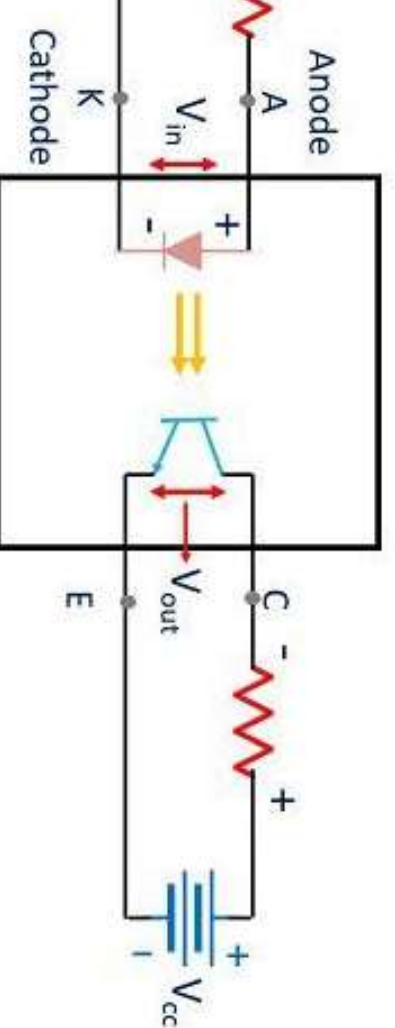
Working of Opto-couplers

voltage appears across the input side of the current start to flow through the LED.

ent LED will emit light.

nt when falls on a phototransistor cause a current the same.

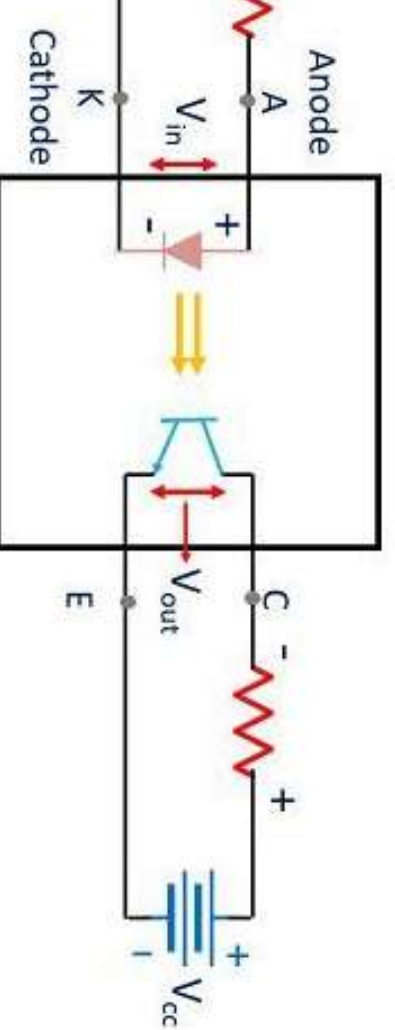
owing through the phototransistor is directly the supplied input voltage.



Working of Opto-couplers

resistor is placed at the beginning of the circuit will control the amount of current flowing through the LED if its value is low so that the current will be low so that the intensity of LED.

When current is flowing through the phototransistor is proportional to the amount of light emitted by the LED.

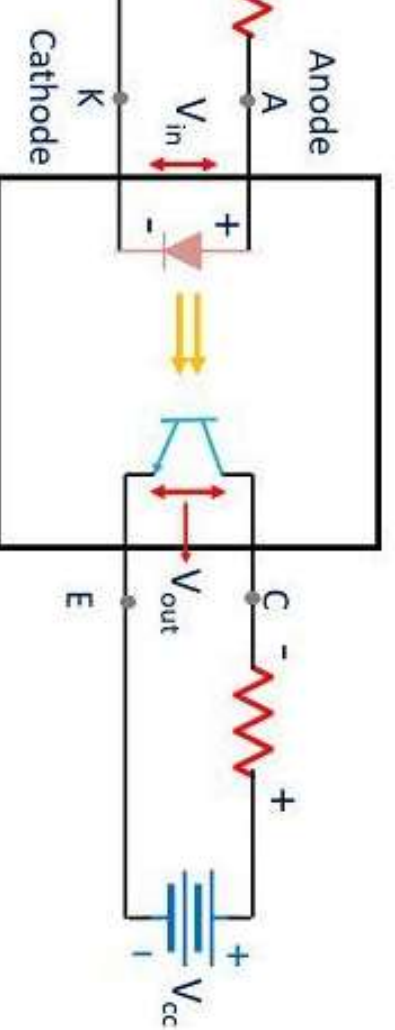


Working of Opto-couplers

At the low-intensity light emitted by the LED will allow current to flow through the phototransistor.

As voltage is generated across the collector-emitter of the transistor.

An incoming signal from the input circuit is coupled to it.



Pulse Transformers

A pulse transformer is one kind of transformer that is designed & used for voltage pulse transmission in between its two

windings. It handles currents & voltages in the pulse form so that it can isolate like Isolating Transformers within power electronic circuits. Its main purpose is to provide a safe source & load.

Advantages include:

- Pulse amplitude can be changed

- Pulse width can be changed

- Isolation Transformer



Pulse Transformers

windings (to minimize flux leakage)

ing capacitance (to ensure that the profile of the
ned on the secondary as cleanly as possible)

e with high frequency signals, the core material
o cope with repeated and rapid magnetization and
n. (Ferrite core)

Pulse Transformers

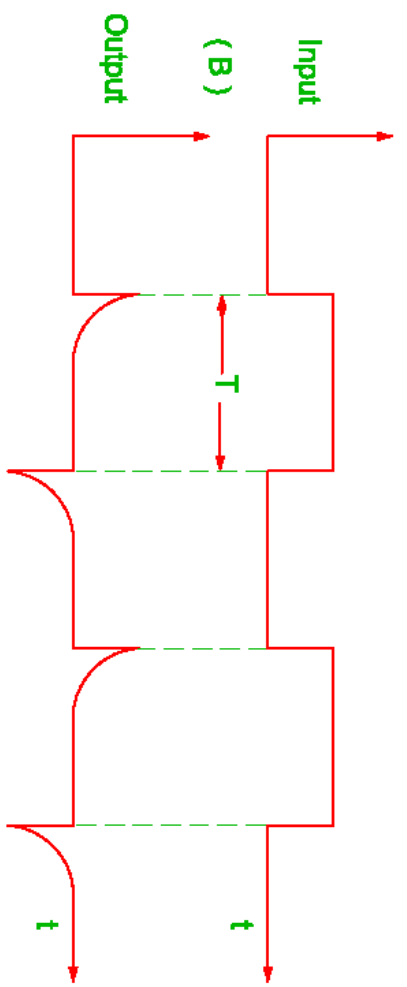
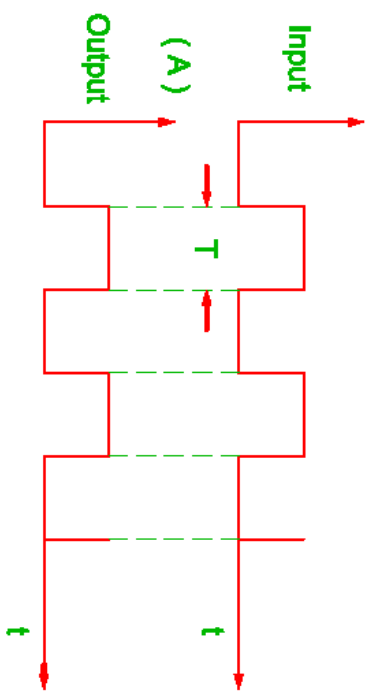
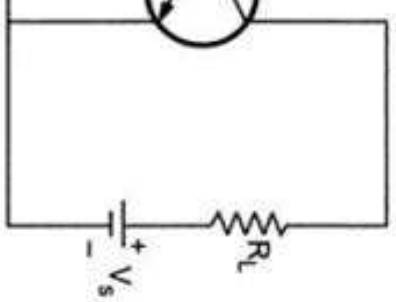


FIG D : Input and Output of Pulse Transformer

Library Assignment - 1

Structure and principle of operation of the
various semiconductor devices.

a

FET

Properties of wideband gap devices- SiC, GaN